

$$\textcircled{1} \lim_{t \rightarrow 0} r(t) = \underbrace{\left\langle \frac{t}{\sin t}, (1+t)^{\frac{1}{t}} \right\rangle}_{\substack{\text{both we know} \\ \text{limits}}}$$

$$\lim_{t \rightarrow 0} r(t) = \left\langle \lim_{t \rightarrow 0} \frac{t}{\sin t}, \lim_{t \rightarrow 0} (1+t)^{\frac{1}{t}} \right\rangle$$

$$\lim_{t \rightarrow 0} r(t) = \left\langle 1, \lim_{t \rightarrow 0} \left(1 + \frac{1}{t}\right)^t \right\rangle$$

$$= \boxed{\langle 1, e \rangle}$$

$$\textcircled{2} \frac{d}{dt} r(t) = \langle \cos t, e^t, \ln t \rangle$$

$$\frac{d}{dt} r(t) = \left\langle \frac{d}{dt} \cos t, \frac{d}{dt} e^t, \frac{d}{dt} \ln t \right\rangle$$

$$\boxed{r'(t) = \langle -\sin t, e^t, \frac{1}{t} \rangle}$$

$$\textcircled{3} \text{ arclen of } r(t) = \langle 5 \cos t, 3 \sin t, 4 \sin t \rangle \text{ on } t=0 \rightarrow t=2\pi$$

$$r'(t) = \langle -5 \sin t, 3 \cos t, 4 \cos t \rangle$$

$$\|r'(t)\| = \sqrt{(-5 \sin t)^2 + (3 \cos t)^2 + (4 \cos t)^2}$$

$$= \sqrt{25 \sin^2 t + 9 \cos^2 t + 16 \cos^2 t}$$

$$= \sqrt{25 \sin^2 t + 25 \cos^2 t}$$

$$= 5\sqrt{1}$$

$$= 5$$

$$\int_0^{2\pi} \|r'(t)\|$$

$$\underbrace{\int_0^{2\pi} \|r'(t)\|}_{\text{arclength}}$$

$$= \int_0^{2\pi} 5$$

$$= [5t]_0^{2\pi} = 5(2\pi) - 5(0)$$

$$= 10\pi - 0$$

$$= 10\pi$$

4. $\underline{a(t) = \langle 0, -32 \rangle}$
 $\underline{v(0) = \langle 10, 50 \rangle}$
 $\underline{r(0) = \langle 0, 0 \rangle}$

$$v(t) = \int a(t) = \langle \int 0, \int -32 \rangle$$

$$v(t) = \langle C, -32t + C \rangle$$

$$v(0) = \langle 10, 50 \rangle$$

$$C = 10 \quad -32t + C = 50$$

$$\underline{v(t) = \langle 10, -32t + 50 \rangle}$$

$$r(t) = \int v(t) = \langle \int 10, \int -32t + 50 \rangle$$

$$r(t) = \langle 10t + C, -16t^2 + 50t + C \rangle$$

$$r(0) = \langle 0, 0 \rangle$$

$$\underline{r(t) = \langle 10t, -16t^2 + 50t \rangle}$$

5. $r(t) = \langle \cos(t), \sin(t), t \rangle$
 find tangent & normal components of a

$$t(t) = \frac{r'(t)}{\|r'(t)\|}$$

$$r'(t) = \langle -\sin(t), \cos(t), 1 \rangle$$

$$\|r'(t)\| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{1 + 1} = \sqrt{2}$$

$$t(t) = \left\langle \frac{-\sin(t)}{\sqrt{2}}, \frac{\cos(t)}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

$$N(t) = \frac{t'(t)}{\|t'(t)\|}$$

$$t'(t) = \left\langle -\frac{\cos(t)}{\sqrt{2}}, -\frac{\sin(t)}{\sqrt{2}}, 0 \right\rangle$$

$$\|t'(t)\| = \sqrt{\frac{\cos^2(t)}{2} + \frac{\sin^2(t)}{2} + 0}$$

$$= \sqrt{\frac{1}{2}} (1)$$

$$= \frac{1}{\sqrt{2}}$$

$$\underline{N(t) = \langle -\cos t, -\sin t, 0 \rangle}$$

$$\textcircled{6} \quad r(t) = \langle t^2, t^2, t^3 \rangle \quad \text{find } a_T(t) \text{ and } a_N(t)$$

$$v(t) = \left\langle \frac{d}{dt} t^2, \frac{d}{dt} t^2, \frac{d}{dt} t^3 \right\rangle \xrightarrow{\frac{v \cdot a}{|v|}} \frac{|v \times a|}{|v|}$$

$$v(t) = \langle 2t, 2t, 3t^2 \rangle$$

$$|v(t)| = \sqrt{8t^2 + 9t^4}$$

$$= t\sqrt{8 + 9t^2}$$

$$a(t) = \langle 2, 2, 6t \rangle$$

$$v(t) \cdot a(t) = 2(2t) + 2(2t) + 6t(3t^2)$$

$$= 8t + 18t^3$$

$$a_T(t) = \frac{t(8 + 18t^2)}{t\sqrt{8 + 9t^2}} = \boxed{\frac{8 + 18t^2}{\sqrt{8 + 9t^2}}}$$

$$\begin{vmatrix} i & j & k \\ 2t & 2t & 3t^2 \\ 2 & 2 & 6t \end{vmatrix}$$

$$v(t) \times a(t) = (12t^2 - 6t^2)i - (12t^2 - 6t^2)j + (4t - 4t)k$$

$$= 6t^2i - 6t^2j + 0k$$

$$= \langle 6t^2, -6t^2, 0 \rangle$$

$$|v(t) \times a(t)| = \sqrt{36t^4 + 36t^4} = t^2\sqrt{72}$$

$$a_N(t) = \frac{t^2\sqrt{72}}{t\sqrt{8 + 9t^2}} = \boxed{\frac{t\sqrt{72}}{\sqrt{8 + 9t^2}}}$$

7. $r(t) = \langle t, t^2, t^3 \rangle$ find curvature κ
 space vector
 valued function $\downarrow$

$$r'(t) = \langle 1, 2t, 3t^2 \rangle$$

$$r''(t) = \langle 0, 2, 6t \rangle$$

$$|r'(t)| = \sqrt{1 + 4t^2 + 9t^4}$$

$$\frac{\|r'(t) \times r''(t)\|}{\|r'(t)\|^3}$$

$$\begin{vmatrix} i & j & k \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix}$$

$$r'(t) \times r''(t) = (12t^2 - 6t^2)i - (6t - 0)j + (2 - 0)k$$

$$= \langle 6t^2, -6t, 2 \rangle$$

9. $f(x, y) = x^3(x^2y^3 - 2y^2)$ find 2nd order partials
 $= x^5y^3 - 2y^2x^3$

$$|r'(t) \times r''(t)| = \sqrt{36t^4 + 36t^2 + 4}$$

$$\kappa = \frac{\sqrt{36t^4 + 36t^2 + 4}}{(\sqrt{1 + 4t^2 + 9t^4})^3}$$

$$= \frac{\sqrt{36t^4 + 36t^2 + 4}}{(1 + 4t^2 + 9t^4)^{3/2}}$$

$$f'(x) = \frac{\partial f}{\partial x} = 5x^4y^3 - 6x^2y^2$$

$$\frac{\partial f}{\partial x} f'(x) = [20x^3y^3 - 12xy^2] \frac{\partial f}{\partial y} f'(y) = [15x^4y^2 - 12x^2y]$$

$$f'(y) = \frac{\partial f}{\partial y} = 3x^5y^2 - 4x^3y$$

$$\frac{\partial f}{\partial x} f'(y) = [15x^4y^2 - 12x^2y] \frac{\partial f}{\partial y} f'(y) = [6x^5y - 4x^3]$$

8. $\lim_{x, y \rightarrow 0, 0} \frac{x^2 - y^2}{x^2 + y^2} = \text{DNE}$

approach along $x=0$

$$\lim_{0, y \rightarrow 0, 0} \frac{0 - y^2}{0 + y^2} = \lim_{y \rightarrow 0} -1 = -1$$

approach along $y=0$

$$\lim_{x, 0 \rightarrow 0, 0} \frac{x^2 - 0}{x^2 + 0} = \lim_{x \rightarrow 0} 1 = 1$$

$\lim_{0, y} \neq \lim_{x, 0}$ so $\lim_{x, y \rightarrow 0, 0} f(x) = \text{DNE} \checkmark$

(10.) $z = x^2y + 3xy^4$ $x = \sin(2t)$ $y = \cos(t)$ $\frac{dz}{dt} = ?$ $t=0$

$$x = \sin(0) \quad y = \cos(0)$$

$$x = 0 \quad y = 1$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = 3(2) + 0(0) = \boxed{6}$$

$$\frac{\partial z}{\partial x} = 2xy + 3y^4 = 2(0) + 3(1)^4 = 3$$

$$\frac{\partial z}{\partial y} = x^2 + 12xy^3 = 0^2 + 12(0)(1)^3 = 0$$

$$\frac{dx}{dt} = 2\cos(t) = 2\cos(0) = 2$$

$$\frac{dy}{dt} = -\sin(t) = -\sin(0) = 0$$

(11.) $f(x,y) = x^2y^3 - 4y$ find the direction at $(2,-1)$ in direction of ∇f

$$\nabla f(x,y) = \langle f_x, f_y \rangle = \langle 2xy^3, 3x^2y^2 - 4 \rangle$$

$$|\nabla f| = \sqrt{4+3} = \sqrt{29}$$

$$f_x(2,-1) = 2(2)(-1)^3 = -4$$

$$f_y(2,-1) = 3(2)^2(-1)^2 - 4 = 8$$

$$u = \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle$$

$$\nabla f(2,-1) = \langle -4, 8 \rangle$$

$$\nabla f \cdot u = -4\left(\frac{2}{\sqrt{29}}\right) + 8\left(\frac{5}{\sqrt{29}}\right) = -\frac{8}{\sqrt{29}} + \frac{40}{\sqrt{29}} = \boxed{\frac{32}{\sqrt{29}}}$$

12. $f(x,y) = xe^y$ at $(2,0)$ what direction is the directional
derivative maximized

$$\nabla f(x,y) = \langle f_x, f_y \rangle = \langle e^y, xe^y \rangle$$

$$\nabla f(2,0) = \langle e^0, 2e^0 \rangle = \langle 1, 2 \rangle$$

$$\|\nabla f(2,0)\| = \sqrt{1+4} = \sqrt{5}$$

$$u = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$